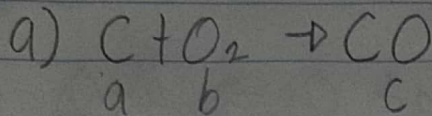
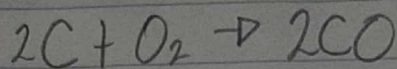


Tarea 10

3.59)



$$C=2$$

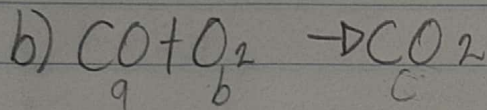


$$C = a = C$$

$$O = 2b = C$$

$$b = \frac{C}{2}$$

$$b = 1$$



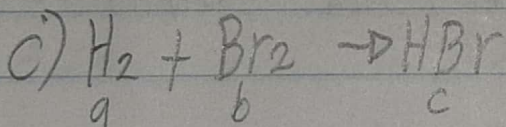
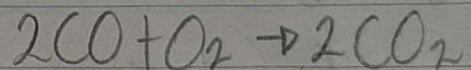
$$C=2$$

$$C = a = C$$

$$O = 2b = 2C$$

$$b = 2C/2$$

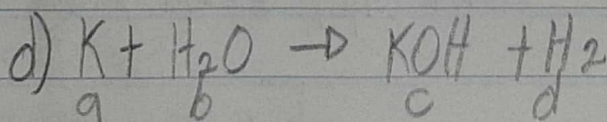
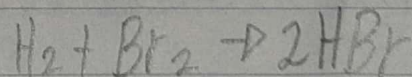
$$b = 2$$



$$C=2$$

$$H = 2a = C = a = 1$$

$$Br = 2b = C$$

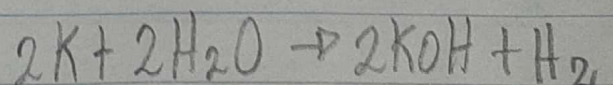


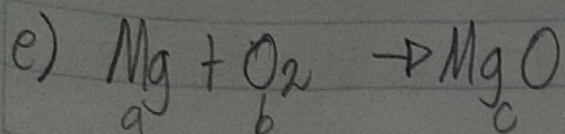
$$C=2$$

$$K = a = C$$

$$H = 2b = 3cd$$

$$O = b = C$$

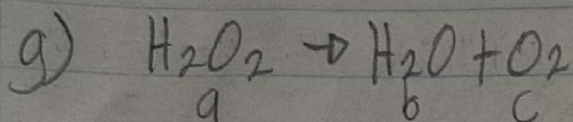
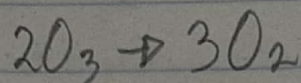
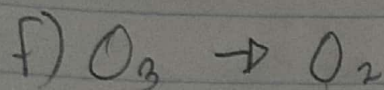
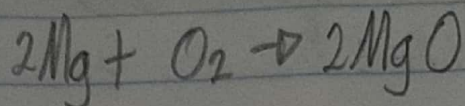




$$c=2$$

$$Mg = a = c$$

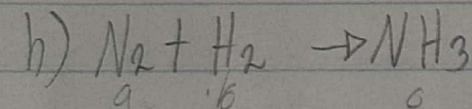
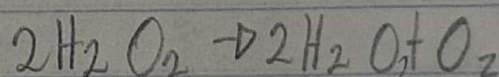
$$O = 2b = c$$



$$b=2$$

$$H = 2a = 2b$$

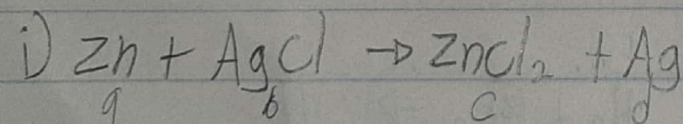
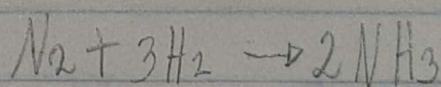
$$O = 2a = 2b + 2c$$



$$c=2$$

$$N = 2a = c$$

$$H = 2b = 3c$$

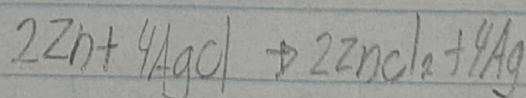


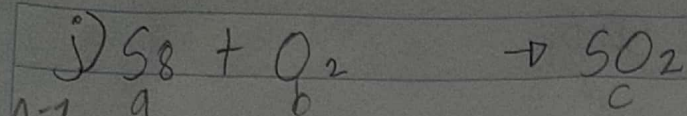
$$c=2$$

$$a=c$$

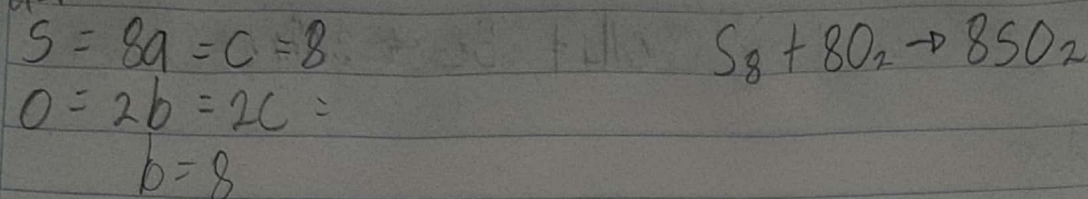
$$b=d$$

$$b=2c$$



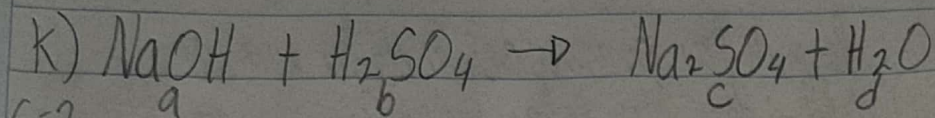
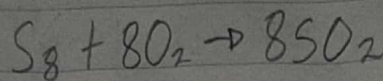


A=1



$$O = 2b = 2c =$$

$$b = 8$$



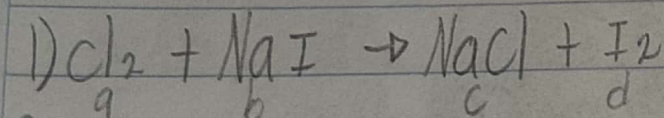
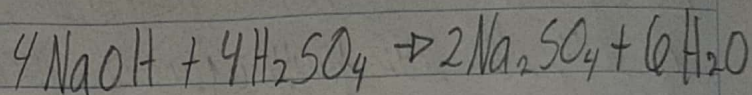
C=2

$$Na: a = 2c = 4$$

$$H: a + 2b = 2d$$

$$S: b = c$$

$$O: a + 4b = 4c$$

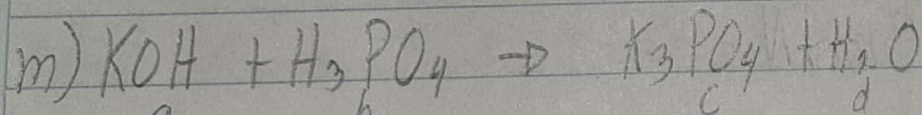
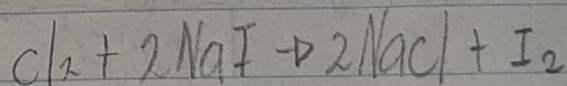


C=2

$$Cl: 2a = c = 1$$

$$Na: b = c$$

$$I: b = 2d$$



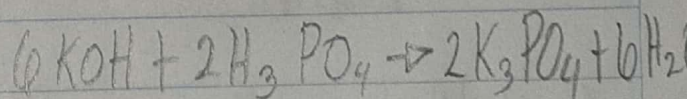
C=2

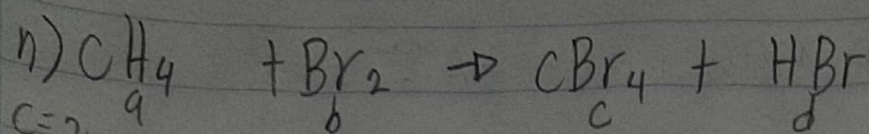
$$K: a = 3c = 6$$

$$H: a + 3b = 2d = 6$$

$$P: b = c = 2$$

$$O: a + 4b = 4c + d$$

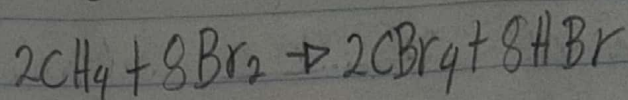




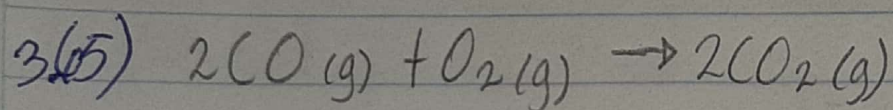
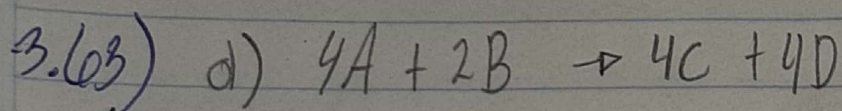
$$\text{C}: a = c$$

$$\text{H}: 4a = d = 8$$

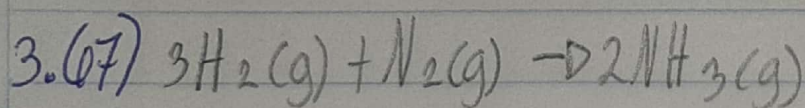
$$\text{Br}: 2b = 4c + d = 8$$



3.61) se basa en 4 leyes: ley de la conservación de la masa, ley de las proporciones definidas, ley de las proporciones múltiples y ley de las proporciones recíprocas

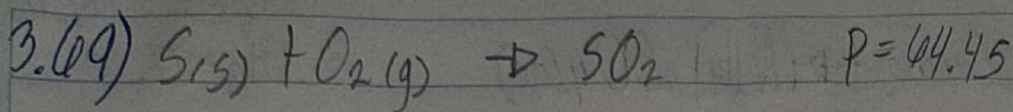


$$3.60 \text{ mol CO} \left| \frac{2 \text{ mol CO}_2}{2 \text{ mol CO}} \right. = 3.60 \text{ mol CO}_2$$



$$6 \text{ mol NH}_3 \left| \frac{1 \text{ mol N}_2}{2 \text{ mol NH}_3} \right. = 3 \text{ mol N}_2$$

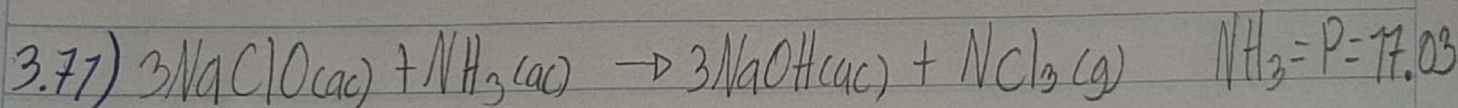
$$6 \text{ mol NH}_3 \left| \frac{3 \text{ mol H}_2}{2 \text{ mol NH}_3} \right. = 9 \text{ mol H}_2$$



$$26,000,000 \text{ ton} \left| \frac{1 \times 10^6 \text{ g}}{1 \text{ ton}} \right| = 2.6 \times 10^{13} \text{ g}$$

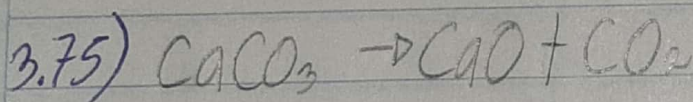
$$2.6 \times 10^{13} \text{ g SO}_2 \left| \frac{1 \text{ mol SO}_2}{64.45 \text{ g SO}_2} \right| \left| \frac{1 \text{ mol S}}{1 \text{ mol SO}_2} \right| \left| \frac{32 \text{ g S}}{1 \text{ mol S}} \right| = 1.29 \times 10^{13} \text{ g S}$$

$$1.29 \times 10^{13} \text{ g} \left| \frac{1 \text{ ton}}{1 \times 10^6 \text{ g}} \right| = 12,900,000 \text{ ton S}$$



$$2.94 \text{ g NH}_3 \left| \frac{1 \text{ mol NH}_3}{17.03 \text{ g NH}_3} \right| \left| \frac{1 \text{ mol NCl}_3}{1 \text{ mol NH}_3} \right| \left| \frac{120.36 \text{ g NCl}_3}{1 \text{ mol NCl}_3} \right| = 20.78 \text{ g NCl}_3$$

$$3.73) (15.01 - 9.6) \text{ g H}_2\text{O} \left| \frac{1 \text{ mol H}_2\text{O}}{18.01 \text{ g H}_2\text{O}} \right| = 0.30 \text{ mol H}_2\text{O}$$

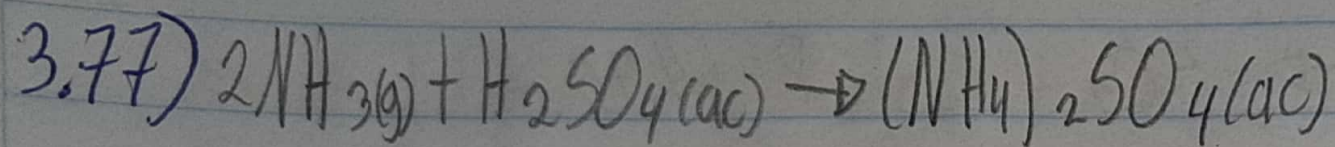


$$\text{CaCO}_3 = 84.09$$

$$\text{CaO} = 56.08$$

$$1 \text{ kg} \left| \frac{1,000 \text{ g}}{1 \text{ kg}} \right| = 1,000 \text{ g}$$

$$1,000 \text{ g CaCO}_3 \left| \frac{1 \text{ mol CaCO}_3}{84.09 \text{ g CaCO}_3} \right| \left| \frac{1 \text{ mol CaO}}{1 \text{ mol CaCO}_3} \right| \left| \frac{56.08 \text{ g CaO}}{1 \text{ mol CaO}} \right| = 666.90 \text{ g CaO}$$



$$(\text{NH}_4)_2\text{SO}_4 = 114.10$$

$$\text{NH}_3 = 17.03$$

$$1 \times 10^5 \text{ Kg } (\text{NH}_4)_2\text{SO}_4 \left| \frac{1 \text{ mol } (\text{NH}_4)_2\text{SO}_4}{114.10 \text{ g } (\text{NH}_4)_2\text{SO}_4} \right| \frac{2 \text{ mol } \text{NH}_3}{1 \text{ mol } (\text{NH}_4)_2\text{SO}_4} \left| \frac{17.03 \text{ g } \text{NH}_3}{1 \text{ mol } \text{NH}_3} \right| = 29,851 \text{ Kg } \text{NH}_3$$